Lecturer:	**Algebra I**	**Due date:**
Pedro Núñez	Sheet 3	15 October 2026

Exercise 3.1. Let V be a finite-dimensional vector space.

(a) Define the natural map $V \to V^{\vee\vee}$.

(b) Prove that this map is an isomorphism.

Solution. For $v \in V$, define

$$\Phi(v)\colon V^{\vee} \to k, \qquad \Phi(v)(\lambda) = \lambda(v).$$

This gives a linear map $\Phi\colon V \to V^{\vee\vee}$, which is the map asked for in Item (a). After choosing a basis of V, the map Φ is identified with the usual identification of a finite-dimensional vector space with its double dual. Hence Φ is an isomorphism, proving Item (b). ■

Exercise 3.2. Let

$$0 \to A \to B \to C \to 0$$

be a short exact sequence of finite-dimensional vector spaces. Use Item (b) of Exercise 3.1 to describe the dual exact sequence.

Solution. Dualizing gives an exact sequence

$$0 \to C^{\vee} \to B^{\vee} \to A^{\vee} \to 0.$$

The finite-dimensionality assumption allows us to identify each vector space with its double dual, using Item (b) of Exercise 3.1. ■